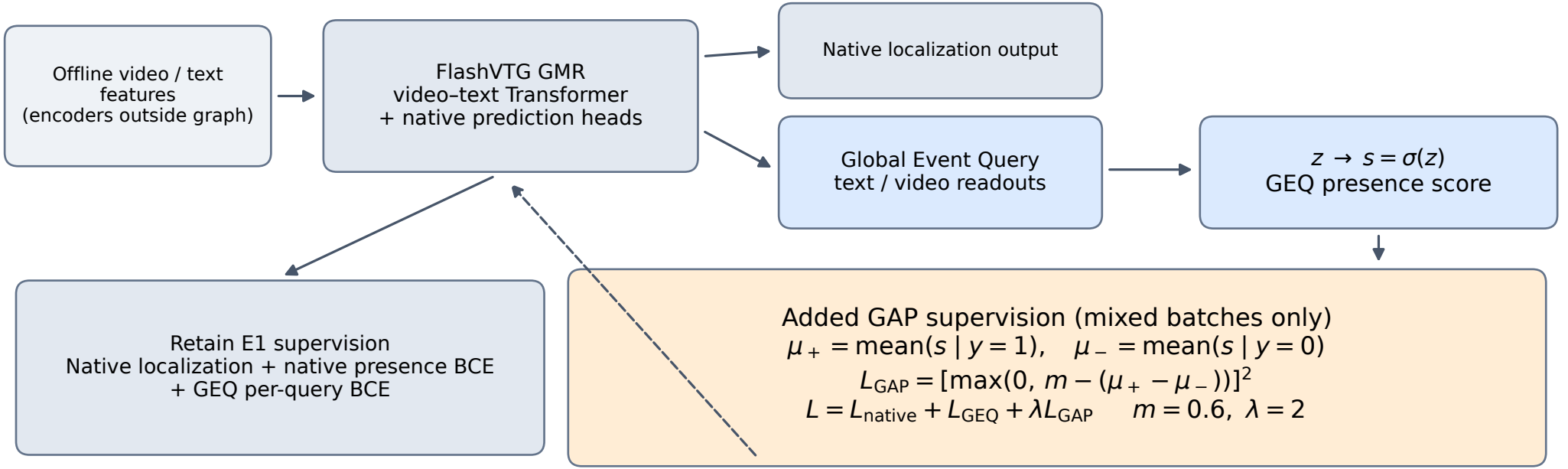
FlashVTG + GEQ + GAP: joint supervision and class-balanced evaluation

Training: original pretrained GMR initialization; all GMR + GEQ parameters jointly optimized



Inference
GEQ alone supplies video-level presence score.
Keep native localization, soft calibration and NMS.
No added gate, PV, boundary refiner or candidate fusion.

Evaluation (positive / negative classes weighted equally)
$D_+ = \mathbb{E}[1 - s \mid y=1], \quad D_- = \mathbb{E}[s \mid y=0]$
$D = (D_+ + D_-)/2, \quad S = 100(1 - D)$
$S = 50(1 + \mu_+ - \mu_-)$; also AUROC, AUPRC and localization

Single-class batches: graph-connected zero GAP; native + GEQ losses remain active. Selection: epoch4 Val AUROC; threshold fixed at 0.4.